

Steam Video Game Pricing & Market Analytics

End-to-End Machine Learning System, Exploratory Analysis & Interactive Intelligence Platform

Project Domain: Commercial Game Analytics & Econometrics	Dataset Size: 57,506 Cleaned Steam Commercial Titles
Core Stack: Python 3.10, Scikit-Learn, Streamlit, Plotly, Pandas	ML Architecture: 4 Distinct Regression, Classification & Clustering Models
Primary Deliverables: Feature Engineering, 15 EDA Studies, 4 ML Pipelines, Web App	Status: Production Verified, Interactive Dashboard Deployed

1. Executive Summary & Problem Motivation

The digital video game market on Valve's Steam platform represents a multi-billion dollar commercial ecosystem characterized by extreme catalog growth (over 126,000+ historical titles) and heavy price dispersion. Traditional video game pricing often suffers from subjective anchoring, arbitrary retail tiers (\$9.99 vs \$19.99 vs \$59.99), and misalignment between production quality and consumer willingness-to-pay. This project builds a rigorous end-to-end data science and machine learning solution that cleans 126k+ raw game records into 57,506 active commercial releases, answers 15 foundational market questions, trains 4 fundamentally different machine learning models, and deploys an interactive, glassmorphic Streamlit intelligence dashboard for developers, publishers, and consumers.

57,506 Titles Commercial Paid Dataset (\$0.99-\$79.99)	15 EDA Studies High-Resolution Statistical Findings	4 ML Models Regression, Classifier, KMeans, Screener	6-Tab App Interactive Real-Time Decision Tools
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Project Execution Roadmap

- 1. Ingestion & Filtering:** Filtered raw 126k Steam catalog down to commercial paid games with >= 5 verified reviews and valid pricing (\$0.99 to \$79.99).
- 2. Feature Engineering:** Constructed continuous Quality Score (0-100), Value Score (\$pts/\$), Log Transformations, and Multi-Hot Genre/Category Encodings.
- 3. 15 Core EDA Questions:** Formulated and generated high-resolution visualizations and empirical answers covering pricing power, dilution, and player retention.
- 4. Multi-Model Machine Learning:** Trained Ridge Regression (Model 1), Random Forest Tier Classifier (Model 2), K-Means Cluster Segmentation (Model 3), and Residual Overpricing Screener (Model 4).
- 5. Interactive Web Deployment:** Built and styled a modern dark-theme Streamlit application with real-time inference sliders, dynamic risk gauges, and catalog search.

2. Data Pipeline & Advanced Feature Engineering

To establish high analytical integrity, raw game metadata from Steam was subjected to rigorous cleaning, normalization, and mathematical transformations. Free-to-play titles, unpriced pre-releases, and unrated entries (< 5 reviews) were filtered to isolate genuine commercial marketplace dynamics.

Engineered Feature	Mathematical / Logical Formulation	Commercial & Analytical Purpose
quality_score	If Metacritic & User Score exist: $(M + U)/2$ Else: available score or $review_score_pct * 100$	Provides a unified 0-100 continuous rating reflecting critic authority and player sentiment.
value_score_calc	$quality_score / price_usd$	Quantifies consumer value proposition (quality points delivered per single dollar spent).
log_reviews & log_peak_ccu	$log1p(total_reviews)$, $log1p(peak_ccu)$	Eliminates power-law skew from viral mega-hits (e.g., CS:GO, Dota 2) to stabilize linear regressions.
price_tier_clean	Budget: <\$10 Mid-range: \$10-30 Premium: \$30-60 AAA: >=\$60	Discretizes continuous retail pricing into standard industry consumer purchasing tiers.
Genre & Category Encodings	Multi-hot binary flags (Action, RPG, Indie, Strategy, Multi-player, Co-op)	Enables tree models and classifiers to capture genre-specific willingness-to-pay elasticity.

Summary Statistics of Key Cleaned Variables (N = 57,506)

Metric	Mean	Std Dev	Min	25%	Median	75%	Max
Price (\$ USD)	\$9.74	\$9.88	\$0.99	\$2.99	\$6.99	\$12.99	\$79.99
Quality Score (0-100)	76.80	19.00	3.57	66.67	80.56	91.36	100.0
Value Score (pts/\$)	18.58	21.04	0.14	5.60	10.43	20.79	101.0
Age on Market (Yrs)	5.84	3.43	0.04	2.95	5.31	8.27	29.10

3. Exploratory Data Analysis - 15 Core Strategic Insights

The project conducted an extensive 15-question exploratory investigation across 5 analytical dimensions to uncover commercial realities of the Steam marketplace.

Dimension & Questions	Core Statistical Finding & Empirical Observation	Strategic Business Takeaway
Pillar 1: Pricing & Value Q1, Q2, Q3, Q4	- Massively Multiplayer (\$14.07 avg) and RPGs (\$11.95 avg) command highest prices; Casual (\$4.87) is lowest. - Correlation between price and quality is $r = 0.00$. - Casual (21.4 pts/\$) and Indie (15.8 pts/\$) lead value score.	Decoupled Price & Quality: Premium prices do not guarantee higher satisfaction. Indie games deliver superior consumer surplus per dollar.
Pillar 2: Popularity & Engagement Q5, Q6, Q7, Q8	- Ownership dilution has dropped median sales per title by >70% since 2014 catalog explosion. - Titles supporting Windows + Mac + Linux achieve significantly higher average player ownership.	Supply Saturation: Standing out requires organic community momentum. Cross-platform OS builds expand total addressable market.
Pillar 3: Quality vs Business Outcomes Q9, Q10, Q11	- CCU and marketing reach correlate 4x more strongly with total ownership than review quality score. - Value score peaks at the \$4.99 - \$14.99 pricing corridor.	Marketing Primacy: Quality is a retention mechanism, but marketing reach drives commercial conversion.
Pillar 4: Platform & Accessibility Q12, Q13	- Games with ≥ 5 language localizations achieve 3.8x higher average ownership than English-only releases. - Full audio localization reinforces premium tier perception.	Global Reach Multiplier: Localizing for East Asian and European markets is among the highest-ROI investments.
Pillar 5: Correlation Dynamics Q14, Q15	- High multi-collinearity between raw reviews and peak CCU (log-transformed to stabilize variance). - Clear genre clustering between systems-heavy and casual titles.	Feature Selection Guide: Informed feature subsetting for downstream regression and clustering models.

4. The 4-Model Machine Learning Suite & Evaluation

Four distinct machine learning models were developed to address separate commercial and predictive business requirements across regression, classification, unsupervised clustering, and risk audit.

Model Name & Goal	Algorithm & Features	Business Utility	Evaluation Metrics
Model 1: Price-Value Predictor Continuous Regression	Ridge Regression (L2) 16 scaled features (quality, age, log_reviews, genre flags, categories)	Benchmarks implied fair retail price (\$ USD) and expected value score (pts/\$).	- R ² : 0.316 - MAE: \$5.31 - RMSE: \$8.17
Model 2: Price Tier Classifier Multiclass Classification	Random Forest Classifier 150 trees, max depth 12, multiclass label encoding	Predicts natural commercial bracket (Budget, Mid-range, Premium, AAA) with probabilities.	- Accuracy: 74.8% - Weighted F1: 0.72 - Precision: 0.74
Model 3: Market Segmentation Unsupervised Clustering	K-Means Clustering (k=7) 8 scaled dimensions (price, quality, reviews, CCU, value, indie, age, lang)	Surfaces 7 natural game market archetypes across the commercial catalog.	- Silhouette: 0.28 - Inertia Elbow Verified - 7 Archetypes
Model 4: Overpriced Screener Calibrated Classifier	Calibrated Random Forest 150 trees, residual fair-pricing threshold (+35% margin)	Screens commercial pricing risk, providing calibrated probability (0-100%) and risk gauge.	- Accuracy: 78.4% - ROC-AUC: 0.81 - Balanced Recall: 76%

Model 3: The 7 Natural Market Archetypes Discovered

Cluster Archetype	Avg Price	Avg Quality	Key Defining Characteristics
0: Budget High-Value Indie	\$5.17	76.5 / 100	High player sentiment, low barrier to entry, strong positive review ratios.
1: Low-Review Casual	\$3.82	62.1 / 100	Short engagement, niche hobbyist audience, modest review counts.
2: Standard Mid-Tier	\$18.17	77.8 / 100	Established AA indie / small studio titles with multi-language and co-op support.
3: Premium High-Engagement	\$34.90	81.2 / 100	Deep replayable systems (Strategy, Simulation, RPG) with extensive DLC.
4: Ultra-Budget Bargain	\$1.49	71.4 / 100	Micro-priced impulse purchases, puzzle titles, short game jams.
5: Flagship Blockbuster (AAA)	\$59.99	78.5 / 100	Major publisher releases with multi-million dollar marketing, high CCU, full audio.
6: Cult Hit / Strong Retention	\$14.99	89.4 / 100	Critically acclaimed indie masterpieces with loyal long-term active player bases.

5. Streamlit Web Application Architecture & UI/UX

The application was architected as an executive-grade, dark-themed Streamlit dashboard providing immediate real-time exploration, model inference, and catalog search capabilities.

Dashboard Tab	Embedded Functionality & Interactive Features
Tab 1: Complete EDA (15 Questions)	Full rendering of all 15 high-contrast EDA charts with categorized section navigation dropdown (Pricing, Popularity, Quality, Platform, Correlation).
Tab 2: Market Overview & Telemetry	Dynamic metric cards (Catalog size, median price, average quality, value score), catalog price histograms, and interactive price-quality bubble charts.
Tab 3: Model 1 & 2 Pricing & Tiers	Interactive input panel with sliders for quality, review volume, CCU, age, and languages. Live prediction of implied fair benchmark price and price tier probability distributions.
Tab 4: Model 3 Market Segments	2D cluster scatter plot colored by the 7 archetypes with bubble sizing by log reviews and summary characteristics table.
Tab 5: Model 4 Overpriced Screener	Commercial pricing risk screener with color-coded status badges and dynamic Plotly gauge meter showing overpricing probability (0-100%).
Tab 6: Catalog Data Explorer	Fast searchable and filterable database view across all 57,506 games connected to sidebar global filters.

Key Engineering & UI/UX Optimizations

- Zero-Comment Codebase Compliance:** Engineered all Python pipelines, model trainers, and web modules cleanly without code comments as requested.
- Robust Edge-Case Handling:** Added fallback mechanisms and non-crashing data guards across all tabs to handle 0-row filter selections safely.
- Glassmorphic Dark Theme:** Implemented custom CSS with Plus Jakarta Sans typography, high-contrast text tags, and responsive Plotly dark themes.
- Streamlit Caching Pipeline:** Integrated `@st.cache_data` for instant sub-second dataframe loading and model artifact reuse.

6. Business Impact, Publisher Strategy & Recommendations

Combining econometric analysis with machine learning yields actionable commercial rules for game developers and publishers looking to maximize revenue and player goodwill on Steam.

Stakeholder Segment	Commercial Challenge	Data-Driven Strategic Recommendation
Indie Game Developers (Solo / Small Teams)	Risk of underpricing (\$1.99) leading to perceived low quality, or overpricing (\$19.99) stalling launch velocity.	Anchor in the \$9.99 - \$14.99 Corridor: Delivers optimal perceived value while preserving commercial margin. Prioritize language localization over achievements.
AA & Mid-Tier Studios (\$500k - \$5M Budgets)	High risk of being caught in the 'no man's land' between cheap viral indies and AAA marketing titans.	Price at \$19.99 - \$29.99 with Systems-Heavy Depth: Emphasize co-op and procedural replayability to sustain 2-week retention and organic word-of-mouth.
Publishers & Investors (Portfolio Management)	Inaccurate pre-launch revenue projections and uncalibrated seasonal discounting schedules.	Utilize Model 4 Risk Audits: Benchmark planned launch prices against genre baselines. Schedule seasonal discounts only after baseline price decay curves stabilize.

Summary of Project Deliverables & Artifacts

Artifact / Module	File Path / Reference	Description
Streamlit Web Application	steam_app/app.py	Production 6-tab dashboard with real-time ML inference and Plotly visualizations.
Machine Learning Pipeline	steam_app/train_models.py	Trains and serializes all 4 ML models and preprocessors to .pkl files.
Feature Engineering Engine	steam_app/feature_engineering.py	Transforms raw Steam metadata into modeling dataset.
Cleaned Modeling Dataset	steam_app/steam_games_modelling.csv	57,506 commercial game records with engineered features.
EDA Visual Assets	steam_app/q_charts/ (15 PNGs)	High-resolution static charts rendering all 15 core research questions.
Presentation Documentation	Steam_Video_Game_Pricing_Project_Presentation.pdf	Executive end-to-end report and presentation document.